

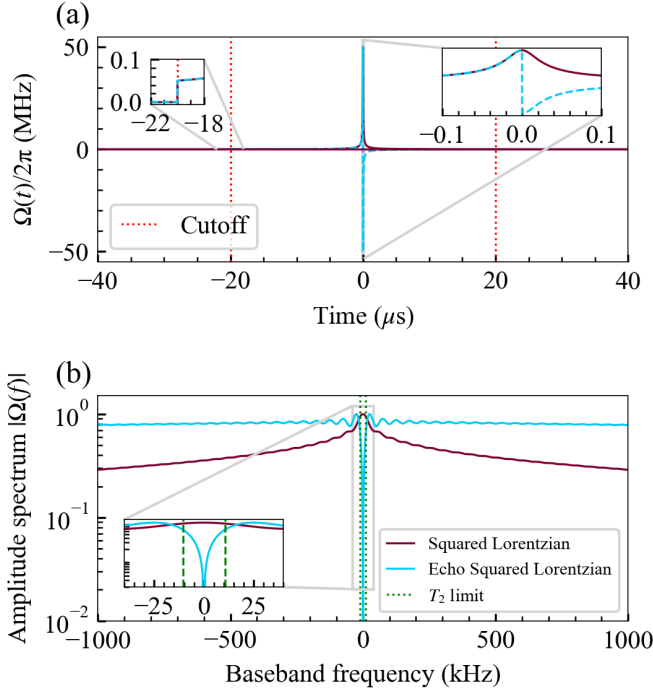


Figure 1. Time and frequency-domain representation of the root-Lorentzian and echo-root-Lorentzian drive pulses. (a) Time-domain profiles of a Lorentzian $L_{1/2}$, and echo-Lorentzian $L_{1/2}^{\text{echo}}$ pulse with a π phase jump at $t = 0$. Red dotted lines indicate the temporal cutoff applied to the pulse envelope. Insets show zoomed-in views of the pulse near the cutoff and phase-jump regions. (b) Corresponding amplitude spectra $|\Omega(f)|$. The echo-root-Lorentzian pulse exhibits a suppression of spectral weight around zero frequency due to destructive interference induced by the phase inversion, resulting in a spectral null at resonance. The T_2 -limited linewidth is indicated by the green dotted line.

mented quantum operation from the ideal one. Minimizing these detuning-induced errors is essential to meet the fault-tolerance thresholds required for scalable quantum computation. [11]

Here, we propose a spectroscopic method for probing the qubit transition frequency using arbitrarily shaped drive pulses, specifically an echo-root-Lorentzian pulse. In previous works by Vitanov [12], it was shown that Lorentzian pulses exhibit power-narrowing, complementary to the conventional power broadening observed for constant drives, originating from the adiabatic properties of these pulses. In this work, we extend the analysis of Vitanov to longer pulse durations approaching the T_1 timescale, thereby entering a regime beyond the Fourier limit. The echo property arises from a π -phase jump applied at the peak amplitude of the pulse, which effectively cancels its net area, while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus

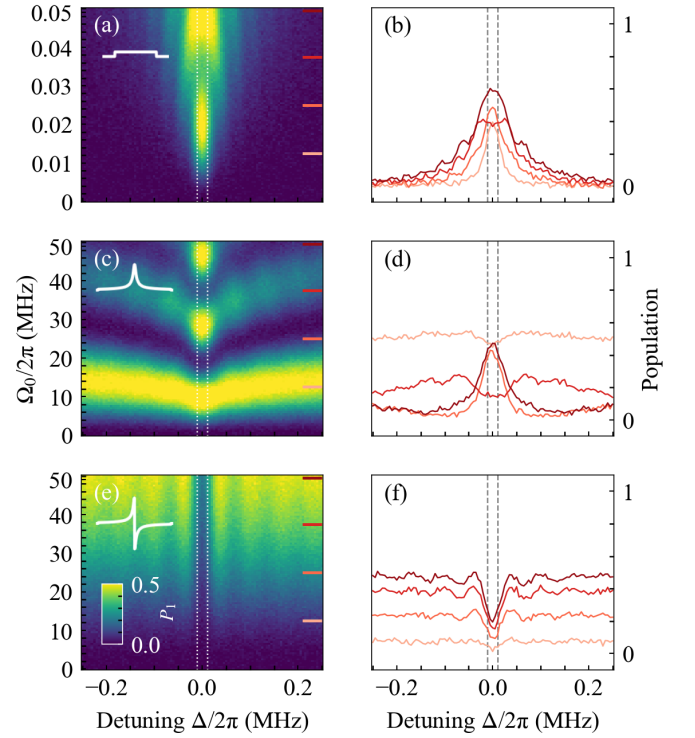


Figure 2. Two-dimensional spectroscopy of the excited-state population as a function of detuning Δ and drive amplitude Ω_0 for three pulse types: (a) square, (c) root-Lorentzian, and (e) echo-root-Lorentzian pulses. Panels (b), (d), and (f) show detuning line cuts at selected amplitudes $\Omega_0/2\pi$, as indicated in the corresponding colored ticks on the left column plots. Vertical dashed lines denote the T_2 limit line width. The square pulse exhibits strong power broadening already at small amplitudes ($\Omega_0/2\pi \sim 0.05$ MHz), whereas the root-Lorentzian and echo-type pulse (30 μs) maintain narrow, T_2 -limited linewidths. The echo-type pulse further suppresses coherent vertical oscillations, enabling smooth high-resolution spectroscopy all over the the amplitude range, but with an expense of SNR.

making resonance distinguishable.

We define the root-echo-Lorentzian $L_{\text{echo}}^{(1/2)}$ drive envelope base on ordered Lorentzian $L^{(1/2)}$ as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \quad (1)$$

$$L_{\text{echo}}^{(1/2)}(t) := L^{(1/2)}(t) \text{sgn}(t). \quad (2)$$

where τ sets the characteristic pulse duration and n controls the temporal sharpness.

Under the adiabatic condition for the plain Lorentzian pulse, the characteristic spectral width obeys the scaling

$$\Delta_{1/2}\tau \propto (\Omega_0\tau)^{-\frac{1}{2n-1}}, \quad (3)$$

where $\Delta_{1/2}$ denotes the typical linewidth and Ω_0 is the peak Rabi frequency. This relation indicates that increasing the drive amplitude suppresses the detuning at which